ETALE NEIGHBOURHOOD SPLITTING A FINITE GROUPOID PIECE OFF A QUASI-FINITE POINT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Quasi-finiteness at the points of a finite fibre (morphisms chapter, Lemma morphisms-lemma-finite-fibre, canonical tag 02NG). cited at proof lines 2450-2452. The structure map s is quasi-finite at each of the finitely many points of the source fibre of u.
- Quasi-finiteness is preserved under composition (morphisms chapter, Lemma morphisms-lemma-composition-quasi-finite, canonical tag 01TL). cited at proof lines 2453-2454. The composite structure morphism from R to S is quasi-finite at each r_i.
- Characterization of quasi-finite at a point by isolation in the fibre (morphisms chapter, Lemma morphisms-lemma-quasi-finite-at-point-characterize, canonical tag 01TH). cited at proof lines 2455-2456 and 2468-2470. Isolates each r_i in the fibre of R over p and, together with openness of the target map of fibres, isolates each u_j in the fibre of U over p, giving quasi-finiteness of the structure morphism at the u_j.
- Flatness is preserved under base change (morphisms chapter, Lemma morphisms-lemma-base-change-flat, canonical tag 01U9). cited at proof lines 2460-2464. The target map of the fibres over p is flat.
- Being locally of finite presentation is preserved under base change (morphisms chapter, Lemma morphisms-lemma-base-change-finite-presentation, canonical tag 01TS). cited at proof lines 2460-2464. The target map of the fibres over p is locally of finite presentation.
- Flat morphisms locally of finite presentation are open (morphisms chapter, Lemma morphisms-lemma-fppf-open, canonical tag 01UA). cited at proof lines 2465-2466. The target map of fibres is open, which propagates isolation from the r_i to the points u_j = t(r_i).
- Finiteness of scheme-theoretic fibres with finitely many points for locally finite type morphisms, much more general form (morphisms chapter, Lemma morphisms-lemma-locally-quasi-finite-qc-source-universally-bounded, canonical tag 03JA). cited at proof lines 2474-2478. Supports that the scheme-theoretic source fibre F_u is finite over the residue field of u.

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- Scheme-theoretic source fibres over two points joined by an arrow with common target become isomorphic over a common field extension (more-groupoids chapter, Lemma lemma-two-fibres, no canonical tag in the snapshot; no-tag pointer to the source label line). cited at proof lines 2480-2482. Transfers finiteness of F_u over the residue field of u to finiteness of F_{u_j} over the residue field of u_j, extending the quasi-finite setup from u to every object of its orbit.
- Technical etale splitting of a quasi-finite part: over an etale neighbourhood with residue field identification the base change decomposes into an open and closed finite piece whose fibre maps bijectively onto the prescribed points and an open complement avoiding them (more-morphisms chapter, Lemma more-morphisms-lemma-etale-splits-off-quasi-finite-part-technical, canonical tag 02LN). cited at proof lines 2506-2521. Produces the etale neighbourhood and the decompositions of the base changes into U' disjoint union W and R' disjoint union W' with U' and R' finite over S', whence conclusions (a) through (d) and the fibre level versions of (e) and (f).

Source reference. Original lemma 02NG: sources/stacks-chow/tags.txt, source lines 3479–3479. Cited at proof lines 2450-2452: s is quasi-finite at each point of the finite source fibre of u.

Source reference. Original lemma 01TL: sources/stacks-chow/tags.txt, source lines 2378–2378. Cited at proof lines 2453-2454: the composite to S is quasi-finite at each r_i.

Source reference. Original lemma 01TH: sources/stacks-chow/tags.txt, source lines 2374–2374. Cited at proof lines 2455-2456 and 2468-2470: isolates r_i in the fibre of R over p and the u_j in the fibre of U over p.

Source reference. Original lemma 01U9: sources/stacks-chow/tags.txt, source lines 2402–2402. Cited at proof lines 2460-2464: the target map of the fibres over p is flat.

Source reference. Original lemma 01TS: sources/stacks-chow/tags.txt, source lines 2385–2385. Cited at proof lines 2460-2464: the target map of the fibres over p is locally of finite presentation.

Source reference. Original lemma 01UA: sources/stacks-chow/tags.txt, source lines 2403–2403. Cited at proof lines 2465-2466: the target map of fibres is open, propagating isolation from r_i to u_j.

Source reference. Original lemma 03JA: sources/stacks-chow/tags.txt, source lines 4632–4632. Cited at proof lines 2474-2478: supports finiteness of the scheme-theoretic fibre F_u over the residue field of u.

Source reference. Original lemma lemma-two-fibres: sources/stacks-more-groupoids/pinned-original-source.tex, source lines 761–761. Cited at proof lines 2480-2482: identifies F_u and F_{u_j} over a common field extension, transferring fibre finiteness within the orbit.

Source reference. Original lemma 02LN: sources/stacks-chow/tags.txt, source lines 3414–3414. Cited at proof lines 2506-2521: produces the etale neighbourhood and the decompositions into finite pieces with bijective fibres.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Let $p \in S$ be a point, and let $u \in U$ be a point lying over p. Assume that*

(1) *$U \to S$ is locally of finite type,*
(2) *$U \to S$ is quasi-finite at u,*
(3) *$U \to S$ is separated,*
(4) *$R \to S$ is separated,*
(5) *s, t are flat and locally of finite presentation, and*
(6) *$s^{-1}(\{u\})$ is finite.*

Then there exists an étale neighbourhood $(S', p') \to (S, p)$ with $\kappa(p) = \kappa(p')$ and a base change diagram

$$\begin{array}{ccccc} R' \amalg W' & = & S' \times_S R & \longrightarrow & R \\ & & \downarrow{\scriptstyle t'} \quad \downarrow{\scriptstyle s'} & & \downarrow{\scriptstyle t} \quad \downarrow{\scriptstyle s} \\ U' \amalg W & = & S' \times_S U & \longrightarrow & U \\ & & \downarrow & & \downarrow \\ & & S' & \longrightarrow & S \end{array}$$

where the equal signs are decompositions into open and closed subschemes such that

(a) *there exists a point u' of U' mapping to u in U,*
(b) *the fibre $(U')_{p'}$ equals $t'\big((s')^{-1}(\{u'\})\big)$ set theoretically,*
(c) *the fibre $(R')_{p'}$ equals $(s')^{-1}\big((U')_{p'}\big)$ set theoretically,*
(d) *the schemes U' and R' are finite over S',*
(e) *we have $s'(R') \subset U'$ and $t'(R') \subset U'$,*
(f) *we have $c'(R' \times_{s', U', t'} R') \subset R'$ where c' is the base change of c, and*
(g) *the morphisms s', t', c' determine a groupoid structure by taking the system $(U', R', s'|_{R'}, t'|_{R'}, c'|_{R' \times_{s', U', t'} R'})$.*

Proof. Let us denote $f : U \to S$ the structure morphism of U. By assumption (6) we can write $s^{-1}(\{u\}) = \{r_1, \ldots, r_n\}$. Since this set is finite, we see that s is quasi-finite at each of these finitely many inverse images, see Morphisms, Lemma 02NG. Hence we see that $f \circ s : R \to S$ is quasi-finite at each r_i (Morphisms, Lemma 01TL). Hence r_i is isolated in the fibre R_p, see Morphisms, Lemma 01TH. Write $t(\{r_1, \ldots, r_n\}) = \{u_1, \ldots, u_m\}$. Note that it may happen that $m < n$ and note that $u \in \{u_1, \ldots, u_m\}$. Since t is flat and locally of finite presentation, the morphism of fibres $t_p : R_p \to U_p$ is flat and locally of finite presentation (Morphisms, Lemmas 01U9 and 01TS), hence open (Morphisms, Lemma 01UA). The fact that each r_i is isolated in R_p implies that each $u_j = t(r_i)$ is isolated in U_p. Using Morphisms, Lemma 01TH again, we see that f is quasi-finite at $u_1, \ldots, u_m$.

Denote $F_u = s^{-1}(u)$ and $F_{u_j} = s^{-1}(u_j)$ the scheme theoretic fibres. Note that F_u is finite over $\kappa(u)$ as it is locally of finite type over $\kappa(u)$ with finitely many points (for example it follows from the much more general Morphisms, Lemma 03JA). By Lemma lemma-two-fibres we see that F_u and F_{u_j} become isomorphic over a common field extension of $\kappa(u)$ and $\kappa(u_j)$. Hence we see that F_{u_j} is finite over $\kappa(u_j)$. In particular we see $s^{-1}(\{u_j\})$ is a finite set for each $j = 1, \ldots, m$. Thus we see that assumptions (2) and (6) hold for each u_j also (above we saw that $U \to S$ is quasi-finite at u_j). Hence the argument of the first paragraph applies to each u_j

and we see that $R \to U$ is quasi-finite at each of the points of

$$\{r_1, \ldots, r_N\} = s^{-1}(\{u_1, \ldots, u_m\})$$

Note that $t(\{r_1, \ldots, r_N\}) = \{u_1, \ldots, u_m\}$ and $t^{-1}(\{u_1, \ldots, u_m\}) = \{r_1, \ldots, r_N\}$ since R is a groupoid[1]. Moreover, we have $\mathrm{pr}_0(c^{-1}(\{r_1, \ldots, r_N\})) = \{r_1, \ldots, r_N\}$ and $\mathrm{pr}_1(c^{-1}(\{r_1, \ldots, r_N\})) = \{r_1, \ldots, r_N\}$. Similarly we get $e(\{u_1, \ldots, u_m\}) \subset \{r_1, \ldots, r_N\}$ and $i(\{r_1, \ldots, r_N\}) = \{r_1, \ldots, r_N\}$.

We may apply More on Morphisms, Lemma 02LN to the pairs $(U \to S, \{u_1, \ldots, u_m\})$ and $(R \to S, \{r_1, \ldots, r_N\})$ to get an étale neighbourhood $(S', p') \to (S, p)$ which induces an identification $\kappa(p) = \kappa(p')$ such that $S' \times_S U$ and $S' \times_S R$ decompose as

$$S' \times_S U = U' \amalg W, \quad S' \times_S R = R' \amalg W'$$

with $U' \to S'$ finite and $(U')_{p'}$ mapping bijectively to $\{u_1, \ldots, u_m\}$, and $R' \to S'$ finite and $(R')_{p'}$ mapping bijectively to $\{r_1, \ldots, r_N\}$. Moreover, no point of $W_{p'}$ (resp. $(W')_{p'}$) maps to any of the points u_j (resp. r_i). At this point (a), (b), (c), and (d) of the lemma are satisfied. Moreover, the inclusions of (e) and (f) hold on fibres over p', i.e., $s'((R')_{p'}) \subset (U')_{p'}$, $t'((R')_{p'}) \subset (U')_{p'}$, and $c'((R' \times_{s',U',t'} R')_{p'}) \subset (R')_{p'}$.

We claim that we can replace S' by a Zariski open neighbourhood of p' so that the inclusions of (e) and (f) hold. For example, consider the set $E = (s'|_{R'})^{-1}(W)$. This is open and closed in R' and does not contain any points of R' lying over p'. Since $R' \to S'$ is closed, after replacing S' by $S' \setminus (R' \to S')(E)$ we reach a situation where E is empty. In other words s' maps R' into U'. Note that this property is preserved under further shrinking S'. Similarly, we can arrange it so that t' maps R' into U'. At this point (e) holds. In the same manner, consider the set $E = (c'|_{R' \times_{s',U',t'} R'})^{-1}(W')$. It is open and closed in the scheme $R' \times_{s',U',t'} R'$ which is finite over S', and does not contain any points lying over p'. Hence after replacing S' by $S' \setminus (R' \times_{s',U',t'} R' \to S')(E)$ we reach a situation where E is empty. In other words we obtain the inclusion in (f). We may repeat the argument also with the identity $e' : S' \times_S U \to S' \times_S R$ and the inverse $i' : S' \times_S R \to S' \times_S R$ so that we may assume (after shrinking S' some more) that $(e'|_{U'})^{-1}(W') = \emptyset$ and $(i'|_{R'})^{-1}(W') = \emptyset$.

At this point we see that we may consider the structure

$$(U', R', s'|_{R'}, t'|_{R'}, c'|_{R' \times_{t',U',s'} R'}, e'|_{U'}, i'|_{R'}).$$

The axioms of a groupoid scheme over S' hold because they hold for the groupoid scheme $(S' \times_S U, S' \times_S R, s', t', c', e', i')$. □

[1]Explanation in groupoid language: The original set $\{r_1, \ldots, r_n\}$ was the set of arrows with source u. The set $\{u_1, \ldots, u_m\}$ was the set of objects isomorphic to u. And $\{r_1, \ldots, r_N\}$ is the set of all arrows between all the objects equivalent to u.